DATA MINING

Practical Work Project

The escalating climate crisis represents one of the most pressing existential challenges of the twenty-first century. In order to successfully mitigate climate change effects, accurate and timely predictions are urgently needed as it drives policy changes and informs effective adaptation strategies. However, traditional physics-based climate models struggle to yield strong performance under low spatial resolutions and become computationally demanding at high resolutions. Machine learning models constitute a promising solution for both effective and efficient climate modeling and weather prediction.

In this project, we aim to develop an end to end weather framework that uses **climate data** (temperature, pressure, precipitation, wind speed, etc.) to forecast the 2-meter temperature class. By applying **data mining and machine learning techniques**, we will also explore patterns and relationships between environmental variables at different regions.

The main objectives of this project are to:

- Collect and preprocess climate data
- Apply **supervised learning algorithms** to predict “2-meter temperature” class. Since this variable is continuous, its range will first be divided into three equal-width intervals
- Use **unsupervised learning (clustering)** to identify natural groupings in climate data
- Evaluate model performance using standard metrics.

The project follows a three-phase data mining approach combining analysis, prediction, and clustering.

Step 1: Data Analysis and Preprocessing

- Exploratory Data Analysis
- Data Preprocessing
- Feature Engineering:

  Supervised learning: Each instance will consist of the following features: “2m dewpoint temperature”, “Surface latent heat flux”, “Surface net solar radiation”, “Surface net thermal radiation”, “Surface sensible heat flux”, “Runoff”, “Total evaporation”, “10m u-component of wind”, “10m v-component of wind”, “Surface pressure”, “Total precipitation” alongside the month and year with the target “2-meter temperature” variable of the following month. Training and test sets should be split in a chronological order.

  Unsupervised learning: Each instance will consist of “Surface net thermal radiation” and “2-meter temperature” features for each month of a given year, then additional features will be engineered from these variables to capture long-term climate change trends.

Step 2: Supervised Machine Learning Algorithms

- Implementation of K-Nearest Neighbors (KNN), SVM, Random Forest.
- Ensemble Learning Algorithms: Soft and Hard Voting, AdaBoost
- Evaluate and compare the obtained models

Step 3: Unsupervised Machine Learning (Clustering)

- Implementation of K-Means, DBSCAN, AGNES, and AutoEncoder-Kmeans
- Apply the clustering algorithms on each year
- Evaluate and compare the obtained models
- Analyse the cluster evolution across the decade

Study Area:

Algeria : https://simplemaps.com/gis/country/dz

Study TimeLine:

monthly mean from 2015 to Present

Dataset:

Climate dataset: Reduce the spatial resolution of the map by a factor of four by aggregating neighboring grid points

ERA5-Land **Monthly averaged reanalysis** :
https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land-monthly-means?tab=download